Vol 16 Saturday Absorption — Ch 5 / Ch 6 / Ch 9 v0.x deltas + F50/F51 consolidation (substantive content + cross-links, NOT full prose; for Keeper integration + Sunday Phase 2/3)

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Vol 16 Saturday Absorption — Ch 5 / Ch 6 / Ch 9

Delta-capture of Saturday's findings into my three primary Vol 16 chapters, cross-linked to the source F-notes. Substantive content + integration pointers; not full chapter prose. K231c discipline throughout: (1,1) and dual-ρ framings stay CANDIDATE-LEAD.

Ch 5 — Bergman Kernel Algebra: v0.x deltas

1. Dual-ρ structure (F47/F49). D_IV^5 carries two Weyl vectors, both real, distinct roles:
   - compact ρ_SO(5) = (3/2,1/2) (half-sum of B_2 K-roots) → K-Casimirs, heat-trace K-type decomposition, R(k);
   - conformal ρ (restricted-root half-sum w/ multiplicity, Harish-Chandra c-function; first component = Bergman exponent n_C/rank = 5/2) → FK Plancherel c-function, Bergman-bulk/measure side. The Bergman algebra carries both: spectral (K-Casimir) and Plancherel-measure (conformal). New Ch 5 section: "Two ρ's of the Bergman algebra."
2. (1,1) shift vector — CANDIDATE-LEAD (K231c). (1,1) = ρ_conformal − ρ_SO(5); n_C − N_c = 2 = rank. DOWNGRADED from "substrate identity": it is the difference of two conventions, weighs 0 until a DERIVED mechanism (not relabeled) shows it functions as an operator/generator in H^2. Record as candidate-lead; do NOT promote in the chapter.
3. π^5 Plancherel-measure framing (LOAD-BEARING, Phase 2 setup). The Bergman/Plancherel measure carries π^{n_C} = π^5 (n_C = 5). This is the substrate-architectural home of the π^5 in m_p = 6π^5 m_e and the

~200 CROSS masses (Grace Investigation #3). The substrate-mechanism FORWARD derivation of why π^5 = Plancherel measure is Sunday Phase 2 (mine); Ch 5 v0.x captures the *framing* now: π^5 lives on the conformal-ρ/Plancherel-measure side.

Ch 6 — Casey #14 Restriction Sequence: v0.x deltas

1. [H_B, P_restriction] $\neq$ 0 — gradings COUPLED (F51). The K-type grading (compact H_B) and the restriction grading (Casey #14 SO(5,2)→SO(4,2)→SO(3,1)) do not commute. This substantiates Casey #14 deeper: the restriction sequence is not a passive labeling — it actively mixes the K-type structure.
2. The mechanism: SO(3,1) boosts are noncompact. In so(5,2) = k ⊕ p (k = so(5)⊕so(2)), the 3 Lorentz boosts sit in p, not k; the 3 rotations sit in SO(3) ⊂ SO(5) ⊂ K. The noncompact $p^+ \oplus p^-$ are the K-type raising/lowering operators building H^2 (Harish-Chandra holomorphic discrete series). So P_restriction (involving boosts) shifts K-types → [H_B, P_restriction] = Σ_{$\lambda \neq \mu$}(C(λ)−C(μ))(P_restriction)_{$\lambda\mu$}|V_λ⟩⟨V_μ| ≠ 0. New Ch 6 section: “Boosts couple the K-type and restriction gradings.”
3. Cross-link to F48 muon reframe. The muon edge-term N_c^codim-4 (codim-4 = dim of the restriction, Casey #14) is restriction-grading content, NOT color (muon is a color singlet). Ch 6 carries the restriction-grading reading of the codim-4 exponent.

Ch 9 — Composite Substrate-Mass-Mechanism: v0.x deltas (HEADLINE)

1. COMPOSITE v0.5 RETIRED. Document the retraction plainly: m_μ/m_e = 1575/8 + 81/8 = 207 as an *additive* “bulk + boundary” form double-counts under Reading (a) (F44 — bulk and boundary are two realizations of one H^2 object, not two objects), and the additive split would require orthogonal K-type/restriction gradings, which are coupled (F51). So the additive Composite v0.5 form does not survive. This retires the construction I built Thu–Sat (F9/F32/F45 candidate).
2. T190 single-object is the muon form. m_μ/m_e = T190 = (24/π^2)^{C_2} = 206.761 (0.003%). Single transcendental Weyl-structure object (24 = N_c·|W(B_2)|); no additive split; Reading-(a)-safe; colorless (consistent with the muon being a lepton, F48). Honest precision framing (Cal #35): the load-bearing reasons are *structural* — Reading (a) + coupled gradings force a single object — and T190 *is* a single object. The 0.003% precision (below the Two-Tier floor) is corroboration, not proof; the geometry is load-bearing.
3. Cross-side Direction B mechanism (CROSS observables, Phase 3 setup). The ~200 CROSS masses (Grace Investigation #3) have the form (spectral integer) × (π^5 measure) × m_e — e.g. m_p = 6π^5 m_e, m_Υ = 60π^5 m_e, m_D = 12π^5 m_e. This product structure is exactly Investigation #1’s FK-factorization question at scale: clean factorization (spectral × measure) = Direction A /

Bergman autocorrelation = [H_B, P]=0; irreducible = Direction B. Since the gradings are coupled (F51), the CROSS masses are candidates for irreducible two-point (Direction B) structure — the same mechanism as the CKM N_c^2 (F50). New Ch 9 section: "Cross-side masses: spectral × Plancherel-measure product." Phase 3 systematic derivation is Sunday-onward.

F50 / F51 consolidation (Investigations #1 + #2, for next-session continuity)

- Investigation #1 (FK factorization, F50): the two sectors factor OPPOSITELY. CKM = two-point/Direction B (N_c^2 amplitude = 9/40 = λ exactly; single-trace N_c^1 = 3/40 rejected) — FORWARD-derived. Muon = single-object T190 (additive Composite v0.5 retired). Same kind of question, opposite answers, different spaces (color vs H^2-grading). Refines Elie's "one computation decides both."
- Investigation #2 (grading orthogonality, F51): COUPLED — [H_B, P_restriction] $\neq$ 0 (boosts noncompact, $p^+ \oplus p^-$ shift K-types). Confirms muon → T190. The coupling IS the dual-ρ cross-side structure (Investigation #3 cross-side column).
- Investigation #3 (Grace): dual-ρ is a *second layer* — ~78% of the catalog is ρ-neutral (substrate primitives / Integer-Web, upstream of harmonic analysis); ~22% is ρ-organized (spectral vs Plancherel); ~200 CROSS masses are the (spectral × π^5-measure) set where #1/#2/#3 meet. Grace hands me the CROSS test population for Phase 3.

Next-session setup (Sunday)

- Phase 2 (π^5, mine PRIMARY): substrate-mechanism FORWARD derivation that π^5 = the Plancherel/Bergman measure on the conformal-ρ side (Ch 5 framing → derivation). Load-bearing for the CROSS masses.
- Phase 3 (~200 CROSS masses): systematic Direction-A-vs-B test of the (spectral integer) × (π^5 measure) × m_e set, using F50's factorization machinery + Grace's CROSS list. CKM Direction B (closed) is the template.
- Carry-forward leads (weigh 0): (1,1) derived-mechanism (K231c); R-6 promote/retire (gated on Investigation #1 Direction, now closed B → Grace can resolve); full R(k) theorem (Elie's deferred session, a_k non-closed-form per Toy 4014).

Honest status

Findings captured for posterity; chapter integration is Keeper's fold-in. The headline delta is Ch 9: Composite v0.5 retired → T190 (my own construction, retired by the discipline — the right outcome). All dual-ρ / (1,1) framings held at candidate-lead per K231c. Phase 2/3 set up via the π^5-Plancherel framing + CROSS test population.

— Lyra, Sat 2026-06-06 15:35 EDT. Vol 16 Saturday absorption: Ch 5 (dual-ρ both-real F47/F49; (1,1) candidate-lead K231c; π^5=Plancherel-measure framing for Phase 2); Ch 6 (Casey #14 substantiated by [H_B,P_restriction]≠0; boosts noncompact in p, $p^+ \oplus p^-$ shift K-types; codim-4 restriction-grading not color); Ch 9 HEADLINE (Composite v0.5 additive RETIRED — double-counts under Reading (a) + coupled gradings; muon = single-object T190 $(24/\pi^2)$^{C_2} 0.003%, geometry load-bearing precision corroboration per Cal #35; cross-side Direction B for ~200 CROSS masses (spectral integer $\times$ π^5 measure $\times$ m_e)). F50/F51 consolidated: CKM two-point/B (FORWARD), muon T190, gradings coupled. Phase 2 (π^5 Plancherel derivation) + Phase 3 (CROSS mass Direction-A/B sweep) set up. Keeper folds into chapter files; (1,1)/dual-ρ stay candidate-lead K231c.